

Filesystem Maintenance

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Lectures

- 1 System administration introduction
- 2 Operating System installation
- 3 User management
- 4 Application management
- 5 System monitoring
- 6 **Filesystem Maintenance**
- 7 Network services
- 8 Security and Protection
- 9 Introduction to Public Cloud

Outline

- 1 Introduction
 - Goals
- 2 Filesystems
- 3 Disk Handling
- 4 Backups

Goals

Knowledge

- Filesystems
- Backup tools
- Backup media

Abilities

- Filesystem resizing
- Filesystem verification
- Perform and restore backups

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Filesystems (I) - Windows related FS

Simple FSs

- FAT (FAT16) → DOS
 - Small disks (< 4GB)
 - File names 8+3
- exFAT
 - FAT32 Extension
 - Theoretical capacity of 64ZiB (512TiB real)
- FAT32 (VFAT) → Windows 95
 - Larger disks
 - Long filenames
 - Partial definition of soft-links
 - No owner or file access privileges

Server grade FS

- NTFS → WinNT, XP, Vista, Server, and from Windows 7
 - Integrates ownership and privileges (create, modify, access...)
 - Maps to Windows NT security model

Filesystems (II) - i-node based FS (Linux Native)

- `ext3` → *deprecated*
 - Soft/hard links
 - Access privileges
 - Long filenames
 - Adds journaling in respect to `ext2`
 - Eases error recovery
- `xfs`
 - journaling
 - Dynamic i-node management
 - ACLs
 - Very large disk sizes
 - Filesystem activity log
- `ext4`
 - 64 bits addressing
 - Improvements in journaling
 - Delayed allocation
 - Extents
 - 1 exbibyte (EiB) maximum size

Filesystems (and IV)

- btrfs

- Data consistency
- CoW architecture
- Online filesystem check
- Native RAID support

- zfs

- End-to-End Data Integrity
- Copy-on-Write Architecture
- Self-Healing Scrubs
- Adaptive Caching (ARC)
- Transparent Compression
- Dataset-Level Encryption



Nowadays `ext4` is the recommended choice for desktops and workstations, but not for servers or high availability setups

ZFS vs. BTRFS: Architectural Comparison

ZFS (FreeBSD/Oracle/Linux)

- **Model:** Filesystem on top of block device.
- **Checksums:** Mandatory per-block checksums (data integrity).
- **RAID:** Native RAID-Z, no separate metadata layer.
- **Caching:** Adaptive ARC (RAM cache) optimized for working sets.
- **Encryption:** Full-disk encryption (per dataset).

BTRFS (Linux)

- **Model:** Multi-device filesystem (subvolumes).
- **Checksums:** Optional per-block checksums.
- **RAID:** Internal RAID-1/0/5/6/10; no native erasure coding.
- **Caching:** Standard kernel page cache (no ARC).
- **Snapshots:** Copy-on-Write (COW) with subvolumes.

Journal filesystems

- Journal: disk operation registry
 - Eases the recovery of the FS in case of crash or error
 - Minor overhead in disk operations performance
- Journal outside the buffer cache
 - Journal can be stored in another disk or partition
- Ext3/4, XFS, BTRFS, ZFS have journal

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- 3 Disk Handling**
 - Disk Verification
 - Extending disk space
 - Disk quota management
 - Other Management tasks
- 4 Backups

Disk verification

- Process to check a drive for errors
- Detecting errors early may avoid data loss

Reasons

- Hardware errors
- Power shortage
- Operating system bugs
- Administration errors
 - Incorrect machine shutdown

Never verify a mounted filesystem

- High probability of disk corruption
- Verification access skips the buffer cache and acts directly on the device

Types of Disk verification

Logical verification

- Filesystem metadata
- Directory structure
- Lost data recovery
 - Directory `lost+found`

Physical recovery

- Disk blocks with Input/output errors
- Command: `badblocks`

What to do when a partition gets full

- We can buy another disk and integrate it into the system
- We could try to increase the partition size
 - **Danger of data loss**

Install and configure the new disk

- Create filesystem and decide the mount points
- Transfer the required data to the new partition (onto a temporary mount point)
- Re-Mount the partition to the final destination
 - Update `/etc/fstab`
- Maybe you have to reorganize existing directories
 - `/home` → `/homeA` + `/homeB`
 - `/home` → `/home/students` + `/home/professors`

Exercise

Plan and issue a space refactoring for the following partitions:

- /home
- /var

Disk quota (I)

Quota

Ability to limit the amount of data a user (or user group) is able to use in a filesystem (partition)

Requires

- Support from the filesystem
- Support from the kernel

Quota management (II)

Partition preparation

- Mounted using options 'usrquota' and/or 'grpquota'
- It can be done from /etc/fstab

```
/dev/sda3 /home ext4 defaults,usrquota,grpquota 1 1
```

- quotacheck command to create the quota files

```
quotacheck -v -a -g -u -m  
verbose all group user no-remount
```

- Creates
 - /aquota.user
 - /aquota.group

Quota management (III)

- Quota enabling

```
quotaon -v      -a  -g  -u
           verbose all group user
```

- Activates quota mechanisms, usually from `/etc/init.d/`

- Quota disabling

- `/sbin/quotaoff`

- Quota editing (`edquota`)

```
Disk quotas for user rserral (uid 1000):
Filesystem    blocks      soft      hard      inodes    soft      hard
/dev/sda3      10246    1024716    2024716         0         0         0
```

- Data blocs and i-nodes quota
- It is not possible to edit the used blocks/inodes, but the limits of the quota can be changed

Quota management (and IV)

- Visualize quotas: `quota -v`

```
Disk quotas for user rserral (uid 1000):
```

Filesystem	blocks	quota	limit	grace	files	quota	limit	grace
/dev/sda3	1324716*	1024716	2024716	6days	145	0	0	-

* We are above the quotas, within the "hard" limit!!

- "Grace period"
 - Grace time where the user can reach the hard limit, it only raises warnings
 - If the grace period expires, then the system does not allow to go above the soft limit

Other maintenance tasks

Monitoring

- Free space (`df`)
 - Most systems reserve a (5%) of the space to be exclusively used by root
- Occupied space (`du`)

Synchronization

- Write to disk the modified buffers
 - `sync`
 - Update daemon

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 - Full Backup
 - Incremental Backup
 - Reverse Incremental Backup
 - Differential Backup

Backups

Preliminary considerations

- Data to copy
 - User data (home, mail)
 - Program data (DBs, GIT, web, ...)
 - System configuration
 - Binary?
- Backup frequency
 - Data confidence
 - Data importance
- Backup types
 - Full Backup (all)
 - Incremental Backup (only changes)
 - Reverse Incremental Backup (only changes)
 - Differential Backup

Backups

Physical support

- Floppy, disc, CD, tape, network. . .
- To consider:
 - Cost size ratio
 - Reliability
 - Availability
 - Usability
 - Speed

Copy location

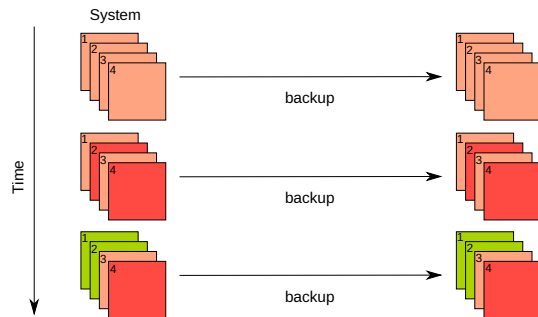
- Accident protection
- Fireproof boxes
- Keep some backups outside the company premises
- Stealing protection

The 3:2:1 Backup Policy

- **3 Copies:** Maintain a total of three data sets.
 - 1 Primary (Production) copy
 - 2 Secondary (Backup) copies
- **2 Media Types:** Store backups on at least two different media types.
 - Examples: Local disk + Cloud storage OR Tape + Network Attached Storage
 - Purpose: Mitigate risk of site-specific failure or media-specific corruption
- **1 Offsite Copy:** Ensure one copy is stored geographically separate.
 - Protects against disasters (fire, flood, theft) at the primary site
 - Essential for Business Continuity Planning (BCP)

Full Backup

- Always copy all the data
 - Fast to restore
 - Large size
- It is the base for all backup types
- Can be compressed
- Can be used on any storage medium



Incremental Backup

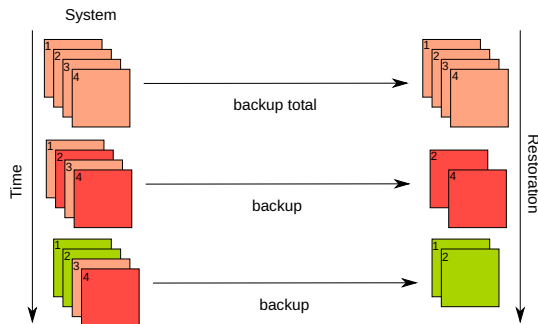
- It only backs up the changed files
- Can be compressed
- Can be used on any storage medium

Advantages

- Smaller size

Inconveniences

- It is slower to restore
- Tedious with long chains



Reverse Incremental Backup

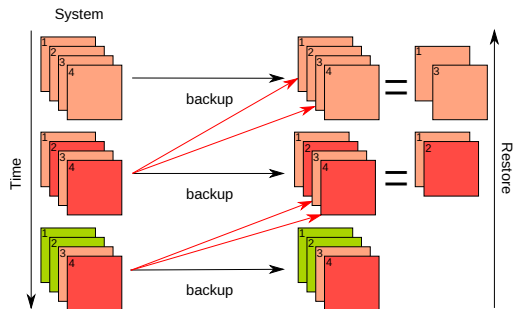
- Everything is copied (linked) from the previous backup
- Except for the modified files
 - Which are taken from the original data

Advantages

- Fast to restore
- Little space

Inconveniences

- Only random access storage



Differential Backup

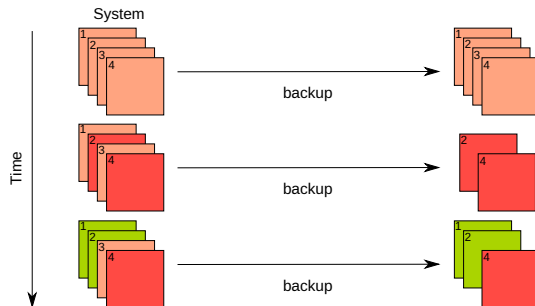
- All changes since the last full backup are copied

Advantages

- Allows to remove Incremental
- Allows a shorter backup chain

Inconveniences

- Large space needs
- Admin should erase incremental



Backups – Tapes¹

	LTO-1	LTO-2	LTO-3	LTO-4	LTO-5	LTO-6	LTO-7	LTO-8	LTO-9	LTO-10	
Release date	2000	2003	2005	2007	2010	2012	2015	2017	2021	2023	2026
Native/raw data capacity	100 GB	200 GB	400 GB	800 GB	1.5 TB	2.5 TB	6.0 TB	12 TB	18 TB	30 TB	40 TB
Advertised capacity (compressed)	200 GB	400 GB	800 GB	1.6 TB	3.0 TB	6.25 TB	15 TB	30 TB	45 TB	75 TB	100 TB
Max uncompressed speed (MB/s)	20	40	80	120	140	160	300	360	400		
Compression capable?	Yes, "2:1"					Yes, "2.5:1"					
Encryption capable?	No			Yes							

¹Source: https://en.wikipedia.org/wiki/Linear_Tape-Open

Exercise

Define a backup policy:

- Data to backup
- Backup type(s)
- Frequency
- Storage Media
- Compression
- Expiration policy

For a server with:

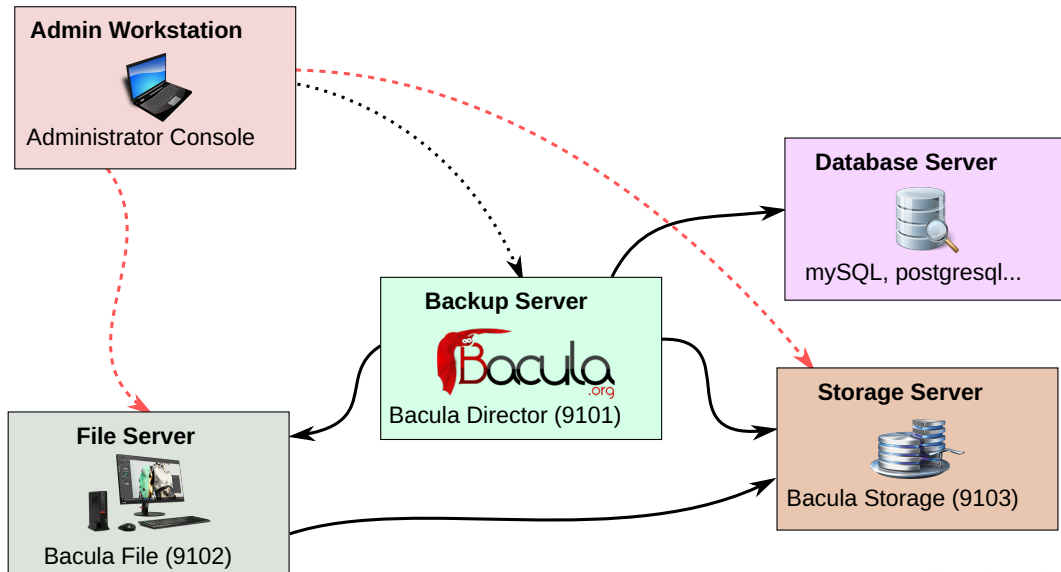
- **Users:** 50 Gb. disk per user (80 users)
- **Mail:** 1 Gb per user
- **Web pages**
 - 2 Gb Main site
 - 100Gb Database
- **Code repository**
 - 100 GB distributed among 10 projects
 - Only 5 active projects

Other considerations

- Dedicate specific machines for backup
 - It takes many resources
 - easier to administer
- Consider how much data you want to potentially loss
- Consider how much storage you will need
- Be aware of legislation

Tools:

bacula, Borg



Personal Homework

- Task automation
 - Programming language: `bash`, `perl`
 - Information search: `find`, `grep`...